IT Project Management — Exam Revision Pack

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Course map: Ch1 Introduction to PM · Ch2 Attributes of a Project · Ch3 System Development Approaches & Project Selection · Ch4 Project Plan Development · Ch5 Developing Budget.
Assessment to expect: MCQs (planning, scope, schedule, cost) + written questions (definitions, triple constraint, PLC/SDLC, WBS, critical path, risk, quality).
CA group topics: Quality Mgmt · Risk · Organizational Change/Conflict · Performance Measurement · Implementation & Closure · Team Building · Communication/Reporting · Working with Management · Managing Teams.

✍ Part 2 — Written Questions: Model Answers

Q1. Define project & project management.

- **Project** — a **temporary endeavour undertaken to accomplish a unique purpose** (PMBOK); it has a definite start and end.
- **Project management** — the application of **knowledge, skills, tools and techniques** to project activities to meet project requirements, balancing the triple constraint (scope, time, cost).

Q2. Main objective of project management?

To **deliver the required scope/deliverables on time, within budget and to the required quality**, satisfying stakeholder expectations — i.e. successfully managing the triple constraint.

Q3. State five areas of the PMBOK.

Any five of the nine: **Integration, Scope, Time, Cost, Quality, Human Resource, Communications, Risk, Procurement** Management.

Q4. Aim of Project Plan Development.

To produce a **usable, flexible, consistent and logical document** that guides the project's work/activities and provides a **control mechanism** for coordinating changes across the project.

Q5. Aim of the Project Planning Process.

To define the agreed **scope, schedule and budget** (the **baseline plan**) — answering what/why/how/who/how long/how much/what can go wrong — used to guide execution and gauge performance.

Q6. Name three subsidiary plans of the project management plan.

Any three: **scope, schedule (time), cost** management plans (also quality, human resource/staffing, communications, risk, procurement plans).

Q7. Why do IT projects fail?

- Lack of **user involvement** and lack of **executive/management support**.
- **Incomplete or changing requirements**; poor/unrealistic planning of schedule and budget.
- Poor communication; rapidly changing technology; difficulty visualizing software and getting accurate requirements.

Q8. Give five attributes of a project.

Any five: **time frame, purpose (unique goal), ownership, resources, roles, risks, assumptions, interdependent tasks, organizational change**, operating in a larger environment.

Q9. Roles of key individuals.

- **Project Manager** — plans, organizes, leads and controls the project; manages scope, schedule, budget and team; the main point of accountability.
- **Project Sponsor** — client/customer/manager who **champions** the project, provides funding, resources and direction, and accepts the final deliverable.
- **Subject Matter Expert (SME)** — a user with specific **knowledge/expertise in a functional area** needed to support the project.
- **Technical Expert** — provides the **technical skills** (hardware, software, networking, development) to design and build the solution.

Q10. Triple Constraint (with diagram) & its effect.

The **Iron Triangle**: **Scope, Time (schedule) and Cost (budget)**, with **Quality** at the centre. The three are interdependent — changing one forces a change in the others (e.g. adding scope increases time and cost; cutting time may raise cost or lower quality).

```
              SCOPE
               /\
              /  \
             / QUALITY \        Change any one side
            /  (centre) \   →   and the others must adjust
           /______________\
        TIME            COST
```

Q11. How and why do projects originate?

From a **business need, problem or opportunity** — e.g. market demand, strategic/organizational need, customer request, technological advance, or legal/regulatory requirement. Management identifies a need that justifies allocating resources, then selects projects on **business value, cost/benefit, technology and risk**.

Q12. Components of the Project Life Cycle (PLC).

(Marchewka, 5 phases): **Define** project goal → **Plan** project → **Execute** project plan → **Close** project → **Evaluate** project.

Q13. Components of the System Development Life Cycle (SDLC).

Planning → Analysis → Design → Implementation (build & test) → Maintenance/Support. Delivered via a structured/waterfall approach or RAD/iterative/spiral approaches.

Q14. Difference between PLC and SDLC.

PLC	SDLC
Management phases of the **whole project** (define, plan, execute, close, evaluate)	Technical phases of **building the system** (planning, analysis, design, implementation, support)
Applies to **any** project; **manages** the project	IT-specific; **produces** the product. Nested inside the PLC's execution phase

Q15. Explain a project's budget.

A function of the project's **tasks/activities, their durations, their sequence and the resources** used (each resource has a cost). It is the total estimated cost of completing all project work; until costs are accounted for, the true cost is unknown.

Q16. Four quantitative (cost-focused) project-selection factors.

- **Payback Period** — time taken for the project's net cash inflows to recover its cost; shorter = better.
- **Return on Investment (ROI)** — (net benefits ÷ costs) × 100; measures profitability relative to cost.
- **Net Present Value (NPV)** — present value of benefits minus costs (discounted for time value of money); positive NPV = worthwhile.
- **Internal Rate of Return (IRR)** — the discount rate at which NPV = 0; compared against the required rate.

Standard cost-focused set (the chapters frame these as "cost/benefit" questions); also commonly cited: breakeven/cost-benefit ratio.

Q17. Project Charter & Project Requirements Document — relevance.

- **Project Charter** — document that **formally authorizes** the project, names the project manager and grants authority, and states objectives, high-level scope and key stakeholders.
- **Requirements Document** — details the **functional and non-functional requirements** (what the product must do).
- **Relevance:** the charter gives the PM legitimacy and authority to start; the requirements document defines the **scope basis** to build the right product and prevent scope creep.

Q18. Which output results from decomposing deliverables into work packages?

The **Create WBS** process output — the **Work Breakdown Structure** (work packages), within Scope Management.

Q19. Define WBS, its benefits, and a model.

WBS = a **deliverable-oriented hierarchical decomposition** of the total project scope into smaller, manageable components (work packages). **Benefits:** organizes/defines the entire scope, makes cost/time estimating easier, assigns responsibility, is the basis for scheduling & control, and prevents missed work.

```
Project
├── 1. Analysis
│     ├── 1.1 Gather requirements
│     └── 1.2 Feasibility study
├── 2. Design
│     ├── 2.1 Database design
│     └── 2.2 UI design
└── 3. Implementation
      ├── 3.1 Coding   (← work package)
      └── 3.2 Testing
```

Q20. Critical Path & its importance.

The **critical path** is the **longest path of dependent activities** through the network; it determines the **shortest possible project duration**. Its activities have **zero slack/float**. **Importance:** any delay to a critical activity delays the **whole project**, so the PM focuses control there and uses it to decide where to compress the schedule.

Q21. Two ways to speed up the schedule.

- **Crashing** — add resources to critical activities to shorten them (**increases cost**).
- **Fast-tracking** — run activities in **parallel** that were planned sequentially (**increases risk** of rework).

Q22. Cost characterization & two estimating approaches.

Cost components: direct vs indirect, fixed vs variable, recurring vs non-recurring. **Two approaches:**

- **Top-down (analogous)** — use overall/historical figures from similar past projects; quick but less accurate.
- **Bottom-up** — estimate each work package/activity and aggregate; more accurate but time-consuming.

Q23. Transform a WBS into a network diagram + draw it.

Take the activities/work packages from the WBS → identify each activity's **predecessors** → draw activities as nodes connected by arrows in dependency order (PDM) → add durations → compute the critical path.

```
Activity  Predecessor   Duration
 A        none (start)    5
 B        A              15
 C        A               5
 D        A              15
 E        C, D           15
 F        B, E            5
 G        F               5

Paths & lengths:
 A-B-F-G       = 5+15+5+5      = 30
 A-C-E-F-G     = 5+5+15+5+5    = 35
 A-D-E-F-G     = 5+15+15+5+5   = 45  ← CRITICAL PATH

CRITICAL PATH = A → D → E → F → G,  Duration = 45 working days
```

Q24. Risk planning, identification, probability & effect, response.

- **Risk planning** — deciding how to approach and conduct risk management for the project.
- **Risk identification** — determining which risks may affect the project and documenting them (in a **risk register**).
- **Risk probability & effect** — assessing each risk's **likelihood** and **impact** to prioritize them.
- **Risk response strategy** — for threats: **Avoid, Transfer, Mitigate, Accept** (for opportunities: exploit, share, enhance, accept).

Q25. What is Quality?

The **degree to which inherent characteristics fulfil requirements** — conformance to requirements and fitness for use;

meeting or exceeding stakeholder needs and expectations.

Q26. Define Project Quality Management.

The processes and activities that determine **quality policies, objectives and responsibilities** so the project satisfies the needs for which it was undertaken — through planning, assuring and controlling quality.

Q27. Quality planning vs assurance vs control.

Process	Focus	What it does
Quality Planning	Plan	Identify quality requirements/standards and how the project will meet them
Quality Assurance (QA)	Process / prevention	Audit the processes to ensure the right processes are used to achieve quality
Quality Control (QC)	Product / inspection	Inspect specific deliverables/results to check they meet standards & find defects

Part 1 — MCQ Answer Key

With brief justification. Q17–19 depend on the PDM figure (not reproduced here); approach shown. Some bank questions are ambiguous — flagged.

#	Ans	Why
1	**D**	Poor planning wastes effort, time AND cost — all of the above
2	**A**	Scope mgmt has the most processes; WBS decomposes work
3	**C**	After charter + stakeholders → develop the project management plan
4	**B**	Holding planning meetings is "meetings", not the expert-judgment technique
5	**D**	Scope mgmt plan covers scope statement, WBS and acceptance — all
6	**A**	Decompose until work is small enough to reliably estimate
7	**A**	Project charter is an input to Collect Requirements
8	**B**	Define Scope needs the requirements documentation
9	**C**	Scope statement documents both work and deliverables
10	**C**	WBS is NOT a tool for work "not yet approved"
11	**D**	Requirements traceability matrix is an output of Collect Requirements
12	**A**	PDM shows four dependency types (FS, FF, SS, SF)
13	**B**	Rolling wave planning
14	**D**	Estimate Activity Resources = type/quantity of material, people, equipment
15	**B**	Parametric (uses a known rate × quantity)
16	**B**	Alternatives analysis is NOT a Develop Schedule tool
17–19	**—**	Need the PDM figure; longest path = critical path, slack = late-start allowed
20	**B**	Summing activity estimates = bottom-up estimating
21	**A**	Define Activities = identify specific actions for deliverables
22	**A**	Analogous (durations of similar past activities)
23	**D**	Resource breakdown structure is NOT an output of Develop Schedule
24	**B**	Parametric (standard cost per square foot)
25	**(?)**	All four are normally inputs to Determine Budget — likely a trick/typo; verify with lecturer
26	**C**	Cost/schedule estimated reliably at the work-package level
27	**A**	Decomposing then aggregating = bottom-up estimating
28	**D**	Determine Budget → project funding requirements

⚡ Quick-Reference Tables

9 PMBOK Knowledge Areas

Integration · Scope · Time · Cost · Quality · Human Resource · Communications · Risk · Procurement.

Estimating techniques

Technique	Use when…
Analogous	You know durations/costs of similar past activities (top-down, quick)
Parametric	You know a standard unit rate (e.g. cost/m², cable/hour)
Bottom-up	You estimate each work package and sum (most accurate)
Three-point	Uncertain — use optimistic, most-likely, pessimistic

System development approaches (Ch3)

Structured / Waterfall — sequential phases, plan up-front, good for stable requirements. **RAD / Iterative / Spiral** — less structured, faster delivery, risk-driven mini-projects/iterations.